

Mantle cell lymphoma

CD5+, cyclin D1+ B-cell lymphoma driven by t(11;14) · a spectrum from observable leukemic non-nodal disease to blastoid disease · **TP53 status, Ki-67 and fitness — not stage — choose the therapy**; BTK inhibitors have displaced routine ASCT

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin — mantle cell (Chou L, Schroers-Martin J, Gupta NK) · NCCN B-Cell Lymphomas v4.2026 · primary trial reports and FDA actions verified against PubMed, September 2026.

DIAGNOSIS

- **Hallmark:** t(11;14)(q13;q32), IGH::CCND1 → **cyclin D1 overexpression** — show it by cyclin D1 IHC or FISH
- **Phenotype:** CD5+, CD20 bright, CD23–, CD10–, cyclin D1+, SOX11+. CD23 negativity and cyclin D1 separate it from CLL, which is CD5+ CD23+
- **Cyclin D1-negative MCL is real:** SOX11+, with a **CCND2 rearrangement in ~55%**. Diagnose it on SOX11 + morphology; treat as MCL *Salaverria, Blood 2013*
- **SOX11 does double duty** — rescues the cyclin D1-negative diagnosis, and its **absence flags the indolent leukemic non-nodal form**

THE SPECTRUM — WHO-HAEM5 / ICC 2022

Form	Features → behavior
Classic (nodal)	SOX11+, IGHV mostly unmutated; nodes + marrow, spleen, GI tract (lymphomatous polyposis). Chronic relapsing course
Leukemic non-nodal	SOX11–, IGHV-mutated, blood + marrow ± spleen, no significant adenopathy. Indolent: 5-yr OS 78% vs 36% for SOX11+ <i>Fernández 2010</i> — observe unless it acquires TP53 or goes nodal
Blastoid / pleomorphic	High Ki-67, TP53 enriched, CNS-prone. Treat as high risk ; blastoid adds nothing to prognosis once Ki-67 is known
In situ mantle cell neoplasia	Cyclin D1+ cells confined to mantle zones of a reactive node — usually incidental; stage and observe

WORKUP

- **PET/CT**, marrow biopsy with flow and t(11;14) FISH, CBC with smear (leukemic phase), LDH, β2-microglobulin, HBV serology before anti-CD20
- **TP53 by NGS (+ del(17p) FISH) and Ki-67 at diagnosis in every patient** — they pick the regimen, not the stage
- **EGD + colonoscopy** for apparent stage I–II (to confirm limited disease) and when GI symptoms
- **LP / brain MRI** if blastoid or neurologic symptoms. Echo before anthracycline

PEARLS

- **CD5+ CD23– small B cells = send cyclin D1.** If cyclin D1– but SOX11+, it is still MCL
- **Send TP53 before choosing.** A TP53-mutant patient gets little from cytarabine + ASCT
- **Rituximab maintenance is standard in MCL** after both intensive and BR induction — unlike follicular, LyMa showed an OS gain at 4 yr (attenuated by 7)
- **No BTKi-added frontline trial in older patients has shown OS gain** (SHINE, ECHO) — PFS gain costs toxicity
- **Brexu-cel is the most active post-BTKi therapy** but the most neurotoxic; liso-cel trades a little efficacy for much less CRS/ICANS

MIPI — ADD THE POINTS

Pts	Age	ECOG	LDH / ULN	WBC ×10 ⁹ /L
0	<50	0–1	<0.67	<6.7
1	50–59	—	0.67–0.99	6.7–9.99
2	60–69	2–4	1.00–1.49	10–14.99
3	≥70	—	≥1.50	≥15

Low 0–3 (44%; median OS not reached) · **intermediate** 4–5 (35%; 51 mo) · **high** 6–11 (21%; 29 mo). *Hoster, Blood 2008* — pre-BTKi survival; use it to stratify, not to counsel.

BIOLOGY THAT OVERRIDES THE SCORE

- **TP53 mutation is the dominant adverse factor.** *11% in Nordic MCL2/3* Median OS **1.8 vs 12.7 yr**, OS HR 6.2; half relapsed within a year of cytarabine induction + ASCT *Eskelund, Blood 2017*
- **Ki-67 ≥30%** is the validated cutoff. **MIPI-c** = MIPI group + Ki-67 → 4 groups, 5-yr OS **85% / 72% / 43% / 17%** *Hoster, JCO 2016*
- **Blastoid/pleomorphic cytology, complex karyotype, del(17p)** travel with TP53 and high Ki-67

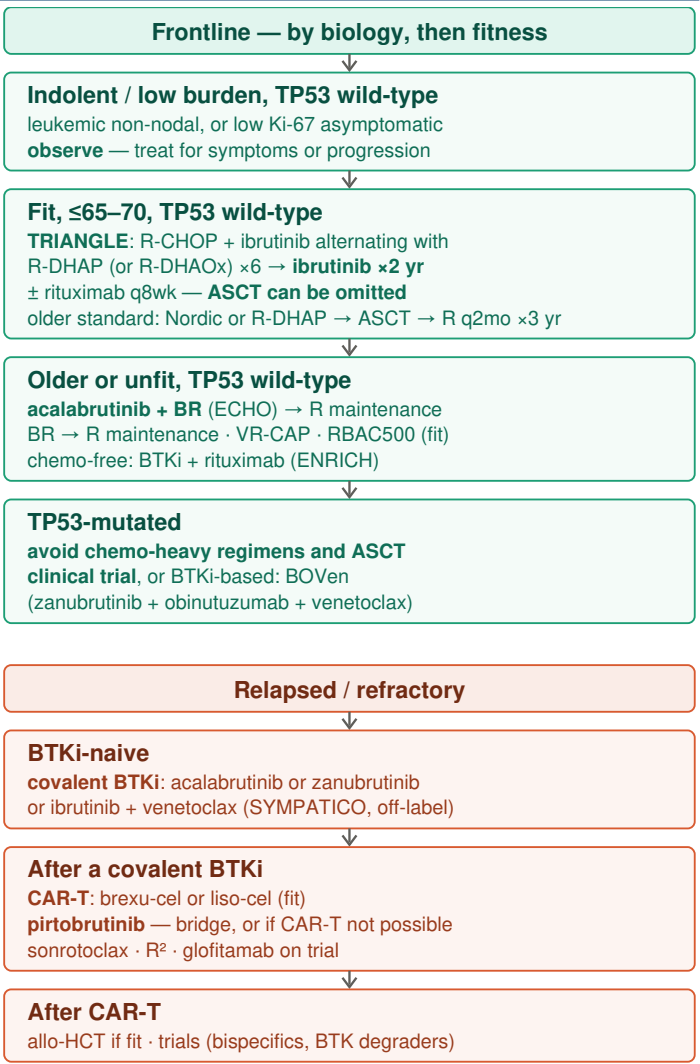
Watch and wait is legitimate for selected patients — leukemic non-nodal SOX11– disease, or asymptomatic classic MCL with low burden, low Ki-67 and wild-type TP53. In a Weill Cornell series **32% were observed**; median time to treatment **12 mo (range 4–128)**, with no survival penalty (a selected cohort) *Martin, JCO 2009*.

ASCT in first remission is no longer default. Its PFS benefit predates BTKi and maintenance. **TRIANGLE:** adding ASCT to ibrutinib gave no FFS gain and more toxicity. **EA4151:** in MRD-negative CR, ASCT + rituximab = rituximab alone (abstract only).

CHEMO-IMMUNOTHERAPY — INFECTION AND SUPPORTIVE CARE

- **Bendamustine is deeply immunosuppressive** — prolonged CD4 lymphopenia: **PJP (TMP-SMX) + VZV (acyclovir) prophylaxis**; ECHO grade 5 COVID-19 **9.4% vs 6.7%**. Rash with allopurinol
- **Rituximab maintenance** → hypogammaglobulinemia and infection; check IgG, replace if recurrent infections. HBV prophylaxis if core Ab+
- **Lenalidomide:** aspirin or anticoagulant VTE prophylaxis, renal dose adjustment

TREATMENT PATHWAY



Mantle cell lymphoma

FRONTLINE EVIDENCE

Trial	Result	How to read it
TRIANGLE <i>≤65, n=870</i>	Ibrutinib + ASCT (A+I) beat ASCT alone (A): 3-yr FFS 88% vs 72% , HR 0.52. At 4.5 yr A+I not better than ibrutinib without ASCT (I) : 4-yr FFS 82% vs 81%. 4-yr OS 88–90% vs 81% for either ibrutinib arm vs A (HR 0.57–0.59)	Primary endpoint is FFS . Maintenance grade 3–5 heme toxicity 54% A+I vs 28% I. Uses ibrutinib, which has no US MCL label ; applying it to acalabrutinib/zanubrutinib is extrapolation
TRIANGLE R-maint <i>secondary</i>	Rituximab maintenance on top of ibrutinib: 4-yr PFS 85% vs 73% (arm I)	Non-randomized; grade 3–5 infections 34% vs 11%
LyMa <i><66, n=240 rand.</i>	R-DHAP → ASCT → rituximab q2mo x3 yr vs observation: 4-yr EFS 79% vs 61% , OS 89% vs 80% (HR 0.50)	PFS gain holds at 7 yr (HR 0.36); OS gain no longer significant
EA4151 <i>MRD– CR1, n=516 rand.</i>	ASCT + R maintenance vs R maintenance alone : 3-yr OS 82.1% vs 82.7% (HR 1.11); PFS 76.6% vs 77.4%. Crossed futility	Abstract only ; few had BTKi induction
Nordic MCL2 <i>single-arm</i>	Maxi-R-CHOP / R-high-dose cytarabine → ASCT: median PFS 8.5 yr , OS 12.7 yr	No plateau — not curative
SHINE <i>≥65, n=523</i>	Ibrutinib + BR → R maintenance: median PFS 80.6 vs 52.9 mo (HR 0.75)	No OS benefit — excess adverse-event deaths; AF 13.9% vs 6.5%
ECHO <i>≥65, n=598</i>	Acalabrutinib + BR: median PFS 66.4 vs 49.6 mo (HR 0.73). Regular FDA approval, Jan 2025	No OS difference (HR 0.86). AF 6.7% vs 4.4%; grade 5 COVID-19 9.4% vs 6.7%
ENRICH <i>≥60, n=397</i>	Ibrutinib + rituximab vs R-chemo: PFS HR 0.69 . vs R-CHOP HR 0.37; vs BR HR 0.91 (NS)	Chemo-free BTKi-R matches BR and beats R-CHOP
VR-CAP <i>n=487</i>	Bortezomib for vincristine in R-CHOP: PFS 24.7 vs 14.4 mo ; median OS 90.7 vs 55.7 mo (HR 0.66)	The one older-patient trial with an OS gain
RBAC500 <i>single-arm, n=57</i>	R + bendamustine + cytarabine 500: CR 91% , median age 71	Missed its safety boundary: 72% dose-reduced , grade 3–4 cytopenias in ~half of cycles
BOven <i>TP53-mut, n=25</i>	Zanubrutinib + obinutuzumab + venetoclax: ORR 96% , CR 88% , 2-yr PFS 72%	Single-arm phase 2 — the best TP53 data, not yet randomized

RELAPSED / REFRACTORY — AGENTS AND STATUS (SEPT 2026)

Agent	Data	FDA status
Acalabrutinib	ACE-LY-004 (n=124): ORR 81% , CR 40%	Accelerated 2017 → regular Jan 2025
Zanubrutinib	BGB-3111-206 (n=86): ORR 84% , CR 69%, median PFS 22.1 mo	Accelerated (Nov 2019), still
Ibrutinib	Pivotal single-arm data; SHINE did not confirm	US MCL indication withdrawn (effective Dec 2023)
Ibrutinib + venetoclax	SYMPATICO (n=267): median PFS 31.9 vs 22.1 mo , HR 0.65 vs ibrutinib	Not approved as a combination in MCL
Pirtobrutinib <i>non-covalent</i>	BRUIN, after covalent BTKi (n=90): ORR 58% , CR 20%, median DOR 21.6 mo; grade ≥3 AF 1.2%	Accelerated (Jan 2023; ≥2 lines incl. BTKi). Confirmatory BRUIN MCL-321 not yet reported
Brexu-cel <i>CD19 CAR-T</i>	ZUMA-2: ORR 93% , CR 67% ; 3-yr median PFS 25.8 mo , OS 46.6 mo. Grade ≥3 CRS 15%, neurologic events 31%	Accelerated 2020 → regular Apr 2026 (added BTKi-naïve cohort: CR 79%)
Liso-cel <i>CD19 CAR-T</i>	TRANSCEND MCL (n=83): ORR 83% , CR 72% , median PFS 15.3 mo. Grade 3–4 CRS 1%, neurologic 9%	Approved May 2024 (≥2 lines incl. BTKi)
Sonrotoclax <i>BCL2</i>	Phase 2 (n=103), post-BTKi: ORR 52% , CR 16%, median PFS 6.5 mo ; TP53-mut ORR 59%; TLS 7%	Accelerated, 13 May 2026 (≥2 lines incl. BTKi)
Glofitamab <i>CD20xCD3</i>	Phase 1/2 (n=60): ORR 85% , CR 78% (71% after BTKi); CRS 70%	Not approved in MCL (US or EU) — phase 3 GLOBRYTE pending
Lenalidomide ± R	SPRINT: PFS 8.7 vs 5.2 mo vs investigator choice (HR 0.61); EMERGE post-bortezomib ORR 28%	Approved 2013 (after bortezomib)
Allo-HCT	Only curative-intent option after CAR-T failure	—

Relapse after frontline BTKi (TRIANGLE, ECHO) is now common: those patients are BTKi-exposed at first relapse, so the post-BTKi column is effectively second line. Response to a covalent BTKi at relapse after BTKi-containing induction is not established.

REGIMEN DOSING

Regimen	Dosing
TRIANGLE (A+I / I)	R-CHOP + ibrutinib : rituximab 375, cyclophosphamide 750, doxorubicin 50, vincristine 1.4 (max 2 mg) d1, prednisone 100 mg d1–5 + ibrutinib 560 mg d1–19 , alternating with R-DHAP (rituximab 375, dexamethasone 40 mg d1–4, cytarabine 2 g/m² q12h x2 d2 , cisplatin 100 CIV d1) or R-DHAOx (oxaliplatin 130) — q21d x6. Then ibrutinib 560 mg daily x2 yr ± rituximab
LyMa	R-DHAP q21d x4 → ASCT → rituximab 375 q2 months x3 yr
BR ± BTKi	Rituximab 375 d1 + bendamustine 90 d1–2 , q28d x6 → rituximab 375 q8wk x2 yr. ECHO adds acalabrutinib 100 mg BID until progression
VR-CAP	Rituximab 375, cyclophosphamide 750, doxorubicin 50 d1; bortezomib 1.3 d1, 4, 8, 11 ; prednisone 100 d1–5 — q21d x6
RBAC500	Rituximab 375 d1, bendamustine 70 d2–3, cytarabine 500 d2–4 — q28d, up to 6
R²	Lenalidomide 20 mg d1–21 q28d + rituximab 375 weekly x4 (cycle 1), then per protocol. VTE prophylaxis
BTK inhibitors	Ibrutinib 560 mg daily · acalabrutinib 100 mg BID · zanubrutinib 160 mg BID or 320 mg daily · pirtobrutinib 200 mg daily — continuous
CAR-T	Brexu-cel : flu 30 + cy 500 mg/m² d–5 to –3, then 2x10 ⁶ CAR-T cells/kg (max 2x10 ⁸). Liso-cel : flu 30 + cy 300 mg/m² x3 d, then 90–110x10 ⁶ CAR+ T cells. Avoid bendamustine shortly before apheresis

mg/m² unless stated. Cytarabine: **prednisolone 1% eye drops** from 12 h before to 48 h after; **reduce to 1 g/m² if age >60**; hold for rising creatinine (cerebellar toxicity).

TARGETED AND CELLULAR THERAPY — TOXICITY

- **BTKi class**: **atrial fibrillation, bleeding, hypertension, infection**, plus arthralgia, diarrhea, rash. Hold **3–7 days around procedures**; avoid warfarin; watch CYP3A interactions
- **Second-generation agents cut AF by about half**: ELEVATE-RR (CLL) **9.4% acalabrutinib vs 16.0% ibrutinib**; ALPINE fewer cardiac events with zanubrutinib; in MCL, ECHO 6.7% vs SHINE 13.9% (cross-trial). Acalabrutinib: early headache. Zanubrutinib: neutropenia. Pirtobrutinib: grade ≥3 AF ~1%
- **CAR-T**: CRS → tocilizumab ± dexamethasone; **ICANS** → dexamethasone (tocilizumab only if CRS coexists), levetiracetam. **Brexu-cel is the more neurotoxic** (grade ≥3 neuro 31% vs 9% liso-cel). Late: cytopenias, hypogammaglobulinemia, boxed warning for T-cell malignancy
- **Venetoclax / sonrotoclax**: **TLS** — MCL can be bulky and leukemic; ramp-up, hydration, allopurinol, lab checks at each step